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| **Fiscal Year:** | FY 2021 | **Task Last Updated:** | FY 08/27/2021 |
| **PI Name:** | Laurie, Steven Ph.D. | | |
| **Project Title:** | Distribution of Body Fluids during Long Duration Space Flight and Subsequent Effects on Intraocular Pressure and Vision Disturbance | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | steven.laurie@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 281-244-0029 |
| **Organization Name:** | KBR/NASA Johnson Space Center | | |
| **PI Address 1:** | Cardiovascular and Vision Laboratory | | |
| **PI Address 2:** | 2400 NASA Pkwy | | |
| **PI Web Page:** | | | |
| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058-2749 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2011 Crew Health NNJ11ZSA002NA |
| **Start Date:** | 08/01/2020 | **End Date:** | 03/31/2022 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Johnston, Smith M.D. ( NASA Johnson Space Center )<br>Lee, Stuart Ph.D. ( KBR/NASA Johnson Space Center )<br>Martin, David M.S. ( KBR/NASA Johnson Space Center )<br>Ploutz-Snyder, Robert Ph.D. ( Universities Space Research Association, Columbia )<br>Smith, Scott Ph.D. ( NASA Johnson Space Center )<br>Soller, Babs Ph.D. ( Reflectance Medical Inc. )<br>Marshall-Goebel, Karina Ph.D. ( KBR/NASA Johnson Space Center )<br>Ribeiro, Laura Christine J.D., M.S. ( KBR/NASA Johnson Space Center )<br>Stenger, Michael ( NASA Johnson Space Center )<br>Dulchavsky, Scott ( Henry Ford Health System )<br>Ebert, Douglas ( KBR/NASA Johnson Space Center )<br>Hargens, Alan ( University of California, San Diego )<br>Macias, Brandon ( NASA Johnson Space Center )<br>Zwart, Sara ( NASA Johnson Space Center )<br>Sargsyan, Ashot ( KBR/NASA Johnson Space Center ) | | |

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| | Arbeille, Phillippe ( University of Tours ) |
| **Grant/Contract No.:** | Internal Project |
| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | NOTE: Continuation of "Distribution of Body Fluids during Long Duration Space Flight and Subsequent Effects on Intraocular Pressure and Vision Disturbance" with new Principal Investigator (PI) Dr. Steven Laurie. Previous PI was Dr. Michael Stenger, who becamse Human Health & Countermeasures Element Scientist.<br>The central objective of the proposed work is to characterize the magnitude of the headward fluid shift during long duration space flight and to measure the subsequent compartmentalization of this fluid. The second objective is to correlate in-flight alterations of eye structure, ocular vascular parameters, and vision with headward fluid shifts, vascular dimensions, and flow patterns. Finally, the third objective is to determine systemic and ocular factors of individual susceptibility to the development of these inflight alterations.<br><br>In order to determine the effect of space flight on the headward fluid shift and subsequent redistribution across fluid compartments, biochemical, ultrasound, and other non-invasive cardiovascular measures will be performed before launch, during flight, and after landing. Total body water, extracellular, and intracellular fluid volume will be determined by biochemical dilution techniques. Ultrasound will be used to assess upper vs. lower body interstitial fluid and vascular flow and structure changes, ocular dimensions and optic nerve sheath diameter, and jugular venous pressure. Near infrared spectroscopy will be used to measure lower vs. upper body local tissue hydration, and total peripheral resistance will be calculated from noninvasively acquired blood pressure and cardiac output.<br><br>It is currently unclear why only some astronauts experience vision disturbances with space flight. It is hypothesized that astronauts respond differently to the fluid shift inflight and that this impacts changes in ocular structure and function changes. The goal of this study is to test this hypothesis. The research proposed here will determine the effect of long duration space flight on fluid shifts and subsequent redistribution across fluid compartments and attempt to discern the contribution of these adaptations to changes in ocular structure and function.<br><br>NOTE: This study was merged with investigations from Dr. Alan Hargens (Fluid distribution before, during and after prolonged space flight) and Dr. Scott Dulchavsky (Microgravity associated compartmental equilibration) resulting in a comprehensive study titled “Fluid Shifts Before, During and After Prolonged Space Flight and Their Association with Intracranial Pressure and Visual Impairment” (short title: Fluid Shifts). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | An early hypothesis considered elevated intracranial pressure (ICP) as a cause for space flight associated neuro-ocular syndrome. Because of this, novel noninvasive ICP techniques included tympanic membrane displacement and otoacoustic emissions are being investigated to determine ICP changes associated with space flight. Knowledge gained from this study may inform medical professionals treating and studying patients suffering from idiopathic intracranial hypertension, a debilitating condition with some characteristics in common with that experienced by astronauts. |
| **Task Progress:** | Within this reporting period the final post-flight data collection was completed on the 13th subject. Data collection is now complete and preparing data reports is underway.<br>Data collected as a part of this project were included in 10 podium and poster presentations at scientific conferences and workshops. In addition, the final 2 crew reports were completed to highlight individual results from those crewmembers' participation in this study.<br><br>Partial results from this investigation were made available during this reporting period with 2 publications in the journal Journal of Applied Physiology and 2 publications in the journal JAMA Ophthalmology, bringing the total publications from this study to 6. Additional manuscripts are still being prepared for submission in the coming months. |
| **Bibliography Type:** | Description: (Last Updated: 12/10/2025) |
| **Articles in Peer-reviewed Journals** | Macias BR, Ferguson CR, Patel N, Gibson C, Samuels BC, Laurie SS, Lee SMC, Ploutz-Snyder R, Kramer L, Mader TH, Brunstetter T, Alferova IV, Hargens AR, Ebert DJ, Dulchavsky SA, Stenger MB. "Changes in the optic nerve head and choroid over 1 year of spaceflight." JAMA Ophthalmol. 2021 Jun;139(6):663-7. https://doi.org/10.1001/jamaophthalmol.2021.0931 ; PMID: 33914020; PMCID: PMC8085766 , Jun-2021 |
| **Articles in Peer-reviewed Journals** | Greenwald SH, Macias BR, Lee SMC, Marshall-Goebel K, Ebert DJ, Liu JHK, Ploutz-Snyder RJ, Alferova IV, Dulchavsky SA, Hargens AR, Stenger MB, Laurie SS. "Intraocular pressure and choroidal thickness respond differently to lower body negative pressure during spaceflight." J Appl Physiol (1985). 2021 Aug 1;131(2):613-20. https://doi.org/10.1152/japplphysiol.01040.2020 ; PMID: 34166098 , Aug-2021 |
| **Articles in Peer-reviewed Journals** | Marshall-Goebel K, Macias BR, Kramer LA, Hasan KM, Ferguson C, Patel N, Ploutz-Snyder RJ, Lee SMC, Ebert D, Sargsyan A, Dulchavsky S, Hargens AR, Stenger MB, Laurie S. "Association of structural changes in the brain and retina after long-duration spaceflight." JAMA Ophthalmol. 2021 Jul 1;139(7):781-4. https://doi.org/10.1001/jamaophthalmol.2021.1400 ; PMID: 34014272; PMCID: PMC8138750 , Jul-2021 |
| **Articles in Peer-reviewed Journals** | Arbeille P, Zuj KA, Macias BR, Ebert DJ, Laurie SS, Sargsyan AE, Martin DS, Lee SMC, Dulchavsky SA, Stenger MB, Hargens AR. "Lower body negative pressure reduces jugular and portal vein volumes, and counteracts the cerebral vein velocity elevation during long-duration spaceflight." J Appl Physiol (1985). 2021 Sep 1;131(3):1080-7. https://doi.org/10.1152/japplphysiol.00231.2021 ; PubMed PMID: 34323592; PubMed Central PMCID: PMC8461809 , Sep-2021 |